SERVICENOW SYSTEM ADMINISTRATOR · NAAN MUDHALVAN (SMARTBRIDGE)

Solution Architecture — Script-Controlled ACL (Institution Details)

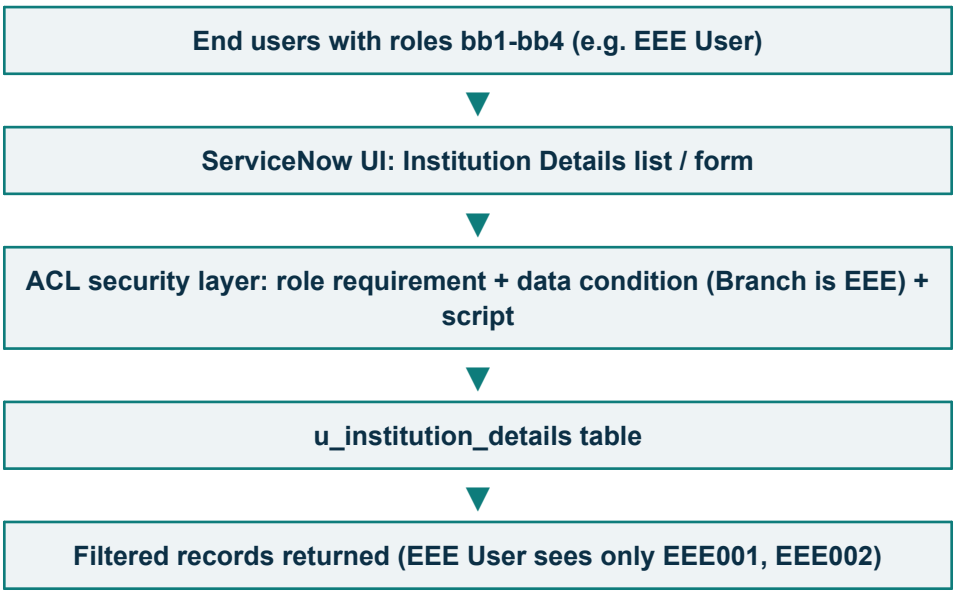
Phase 3 · Project Design Phase

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This document describes the solution architecture for the Script-Controlled ACL project on ServiceNow PDI dev443713. It shows how a request from a branch end-user flows through the ServiceNow UI into the Access Control List security layer, where a role requirement, a data condition and a server-side script together decide access to rows in the u_institution_details table, and how filtered records are returned. All layers are native ServiceNow platform components; there is no external application, framework or database.

Platform	ServiceNow Now Platform (Australia / Zurich family release)
Instance	dev443713.service-now.com (Personal Developer Instance)
Security engine	Access Control List (ACL) with server-side GlideScript (gs.hasRole)
Core table	u_institution_details with Branch choice field (ECE / EEE / CSE)

Architecture Flow



Component Responsibilities

Component	Responsibility
End users and roles (bb1 read, bb2 create, bb3 write, bb4 delete)	Identify the requester and the operation attempted. The test user EEE User holds all four bb roles; admin (System Administrator) always passes; Abel Tuter holds none.
ServiceNow UI / Institution Details list	Presents the list and form for u_institution_details and issues the read, create, write or delete request; renders only what the ACL layer permits, or the "Security constraints prevent access" message when denied.
ACL security layer (sys_security_acl)	For each operation, evaluates the required role, then the data condition, then the script. Read is Advanced with the condition Branch is EEE (u_branch=EEE); Create, Write and Delete check only their bb role.
READ ACL script (gs.hasRole)	Server-side GlideScript: returns true if the user has role admin, true if the user has role bb1, and false otherwise, enforcing the field-value-plus-role decision for reads.
u_institution_details table	Stores the Institution Details records (Student Roll Number, Student Name, Faculty Name, Branch, Email, Phone Number, Description) including ECE001, EEE001, EEE002 and CSE001; returns only rows that satisfy the ACL evaluation.
Supporting configuration (Impersonation, security_admin, default role removal)	security_admin elevation enables editing ACLs; Impersonation verifies each persona's view; removing the auto-generated u_institution_details_user role ensures every ACL depends only on its bb role.